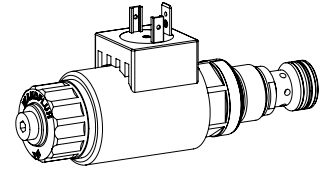


Solenoid operated poppet valve cartridge

- ◆ direct operated
- ◆ 2/2- and 3/2-way
- ◆ $Q_{\max} = 40 \text{ l/min}$
- ◆ $p_{\max} = 350 \text{ bar}$

M22 x 1,5
ISO 7789



DESCRIPTION

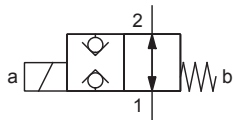
Direct operated 2/2- and 3/2-way solenoid poppet valve in screw-in cartridge construction for cavity according to ISO 7789. By means of the pressure tight switching solenoid, the pressure compensated, metallically sealing poppet spool is either opened or closed. The seat spool guide is sealed by means of an O-ring.

APPLICATION

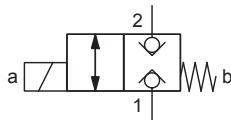
Wandfluh solenoid operated poppet valve cartridges are used where tight closing functions are essential like leakage-free load holding, clamping or gripping. For machining the cartridge cavity in steel and aluminum blocks, cavity tools are available for purchase. Please refer to the data sheets in register 2.13.

SYMBOL

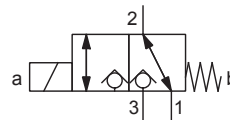
„Normally open“ AB



„Normally closed“ BA



FG



TYPE CODE

Poppet valve		S D S PM22 - - / - #	
Direct operated			
Solenoid			
Screw-in cartridge M22 x 1,5			
Designation of symbols acc. to table			
Nominal voltage U_N	12 VDC G12 24 VDC G24 without coil X5	115 VAC R115 230 VAC R230	
Slip-on coil	Metal housing round with one-sided collar V Metal housing square with one-sided collar N	(only G12 and G24)	
Connection execution	Connector socket EN 175301-803 / ISO 4400 D Connector socket AMP Junior-Timer J Connector Deutsch DT04 - 2P G	(only for $U_N \leq 75 \text{ VDC}$) (only for $U_N \leq 75 \text{ VDC}$)	
Sealing material	NBR FKM (Viton) D1 NBR 872 Z604		
Armature tube	closed 0 with screw plug HB0 with manual override HB4,5		
Design index (subject to change)			

1.11-2061

GENERAL SPECIFICATIONS

Designation	2/2-, 3/2-way poppet valve
Construction	Direct operated
Mounting	Screw-in cartridge construction
Nominal size	M22 x 1,5 according to ISO 7789
Actuation	Switching solenoid
Ambient temperature	-25...+70 °C
Weight	0,56 - 0,60 kg
MTTFd	150 years

ELECTRICAL SPECIFICATIONS

Protection class	Connection execution D: IP65 Connection execution G: IP67 and IP69K Connection execution I/ J: IP66
Relative duty factor	100 % DF
Switching frequency	5'000 / h
Service life time	10 ⁷ (number of switching cycles, theoretically)
Voltage tolerance	± 10 % with regard to nominal voltage
Standard nominal voltage	12 VDC, 24VDC, 115 VAC, 230 VAC AC = 50 to 60 Hz, rectifier integrated in the connector socket

Note! Other electrical specifications see data sheet 1.1-168 (slip-on coil V) and 1.1-175 (slip-on coil N)



SURFACE TREATMENT

- ◆ The cartridge body, the slip-on coil and the armature tube are zinc-nickel coated

HYDRAULIC SPECIFICATIONS

Working pressure	$p_{max} = 350 \text{ bar}$
Maximum volume flow	$Q_{max} = 40 \text{ l/min}$, see characteristics
Leakage oil	Seat tight, max. 0,15 ml / min (approx. 3 drops / min) at 30 cSt
Fluid	Mineral oil, other fluid on request
Viscosity range	12 mm ² /s...320 mm ² /s
Temperature range fluid	-25...+70 °C (NBR) -20...+70 °C (FKM)
Contamination efficiency	Class 20 / 18 / 14
Filtration	Required filtration grade $\beta_{10...16} \geq 75$, see data sheet 1.0-50

ACTUATION

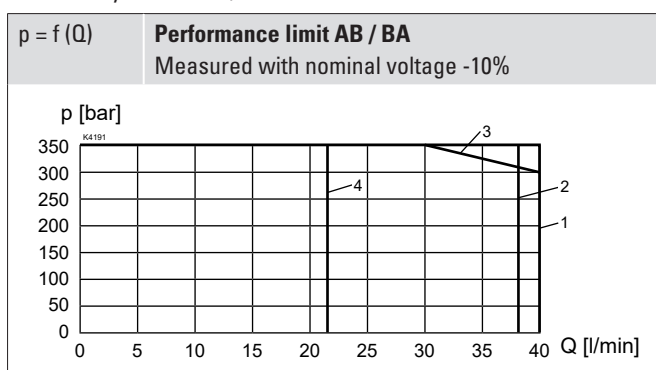
Actuation	Switching solenoid, wet pin push type, pressure tight
Execution	V.E37 / 19 x 50 (data sheet 1.1-168) N.S35 / 19 x 50 (data sheet 1.1-175)
Connection	Connector socket DIN EN 175301 – 803 Connector Deutsch DT04 – 2P Connector socket AMP Junior-Timer axial Connector socket AMP Junior-Timer radial

SEALING MATERIAL

NBR or FKM (Viton) as standard, choice in the type code

PERFORMANCE SPECIFICATIONS

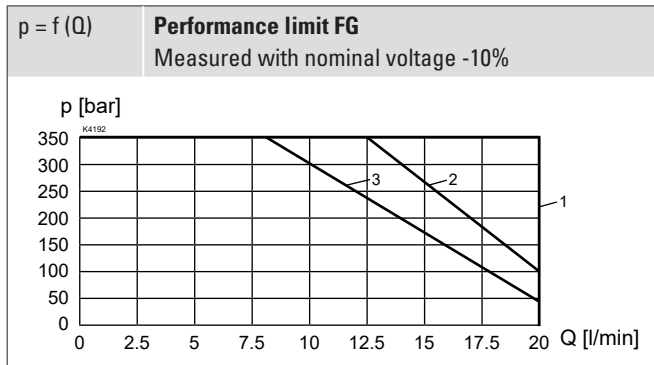
Oil viscosity $\nu = 30 \text{ mm}^2/\text{s}$



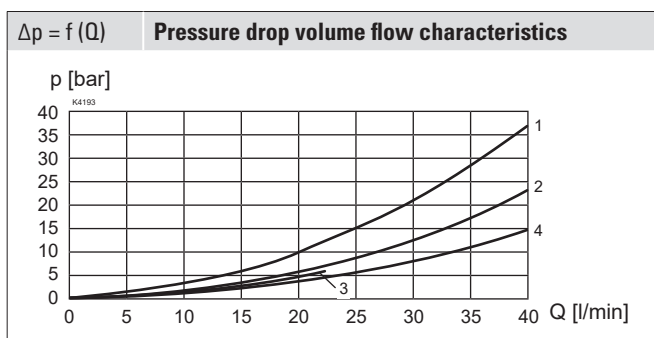
Execution	Flow direction	
	1 → 2	2 → 1
SDSPM22-AB	4	2
SDSPM22-BA	3	1

PERFORMANCE SPECIFICATIONS

Oil viscosity $\nu = 30 \text{ mm}^2/\text{s}$



Execution	Flow direction			
	1 → 2	2 → 1	2 → 3	3 → 2
SDSPM22-FG	2	1	1	3



Execution	Flow direction			
	1 → 2	2 → 1	2 → 3	3 → 2
SDSPM22-AB	3	4	-	-
SDSPM22-BA	2	2	-	-
SDSPM22-FG	3	4	1	1

Switching times

	Type	Flow direction	Switching on	Switching off
SDSPM22	AB	1 → 2	approx. 40 ms	approx. 20 ms
		2 → 1	approx. 40 ms	approx. 10 ms
	BA	1 → 2	approx. 30 ms	approx. 30 ms
		2 → 1	approx. 40 ms	approx. 30 ms
	FG	1 → 2	approx. 40 ms	approx. 10 ms
		2 → 1	approx. 40 ms	approx. 10 ms
		2 → 3	approx. 40 ms	approx. 40 ms
		3 → 2	approx. 40 ms	approx. 20 ms

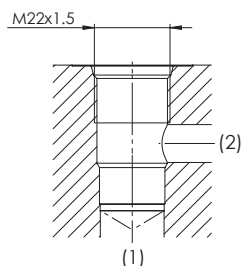
Note!



The switching times depend on the volume flow, pressure and viscosity. In case of very large volume flows, the switching time for closing can get considerably longer.

HYDRAULIC CONNECTION

Cavity drawing according to ISO 7789-22-01-0-98



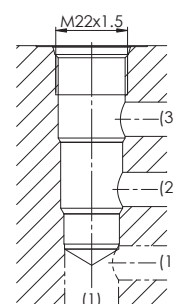
Note!



For detailed cavity drawing and cavity tools see data sheet 2.13-1008

HYDRAULIC CONNECTION

Cavity drawing according to ISO 7789-22-04-0-98



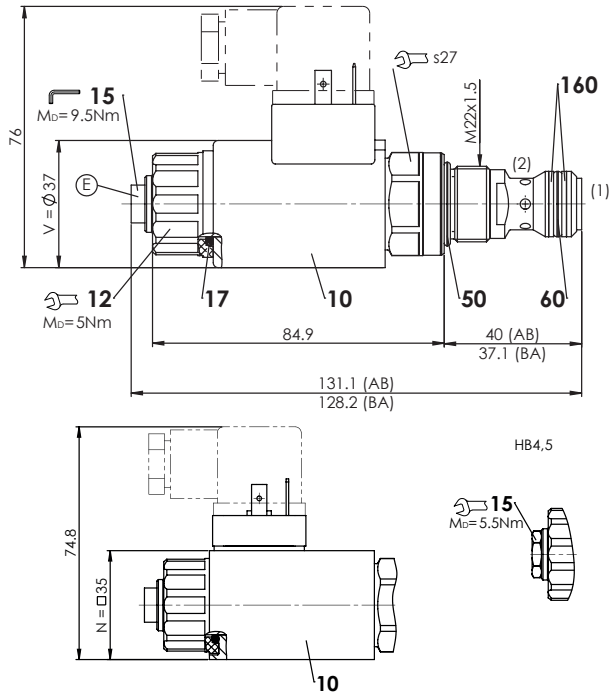
Note!



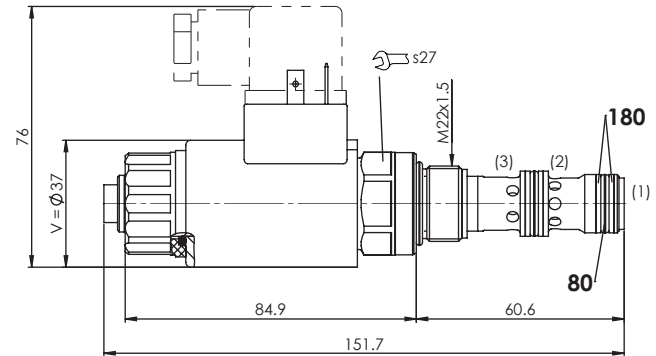
For detailed cavity drawing and cavity tools see data sheet 2.13-1004

DIMENSIONS

SDSPM22-AB / BA



SDSPM22-FG



PARTS LIST

Position	Article	Description
10	206.2...	V.E37 / 19 x 50
	260.5...	N.S35 / 19 x 50
	154.2700	Knurled nut
	253.8000	HB4,5 manual override
	239.2033	HB0 Screw plug
	251.3030	Seal kit SDSPM22-BA-V/N, SDSPM22-AB-V/N
	251.3032	Seal kit SDSPM22-FG-V/N
	-	Seal kit consisting of:
17		O-ring ID 18,72 x 2,62
50		O-ring ID 18,77 x 1,78
60		O-ring ID 15,60 x 1,78
80		O-ring ID 14,00 x 1,78
160		Backup ring rd 16,1 x 19 x 1,4
180		Backup ring rd 14,1 x 17 x 1,4

STANDARDS / TECHNICAL INFORMATION

Cartridge cavity	ISO 7789
Solenoids	DIN VDE 0580
Connection execution D	EN 175301 – 803
Protection class	EN 60 529
Contamination efficiency	ISO 4406

ACCESSORIES

Threaded body	Data sheet 2.9-2xx
Technical explanations	Data sheet 1.0-100
Filtration	Data sheet 1.0-50
Relative duty factor	Data sheet 1.1-430

MANUAL OVERRIDE

Screw plug (HB0): no actuation possible
Optionally: See data sheet 1.1-300 and 1.1-311.

INSTALLATION NOTES

Mounting type	Screw-in cartridge M22 x 1,5
Mounting position	Any, preferably horizontal
Tightening torque	M _D = 60 Nm Screw-in cartridge M _D = 5 Nm knurled nut M _D = 9,5 Nm HB0 M _D = 5,5 Nm HB4,5